ASSIGNMENT-2

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BATCH-16

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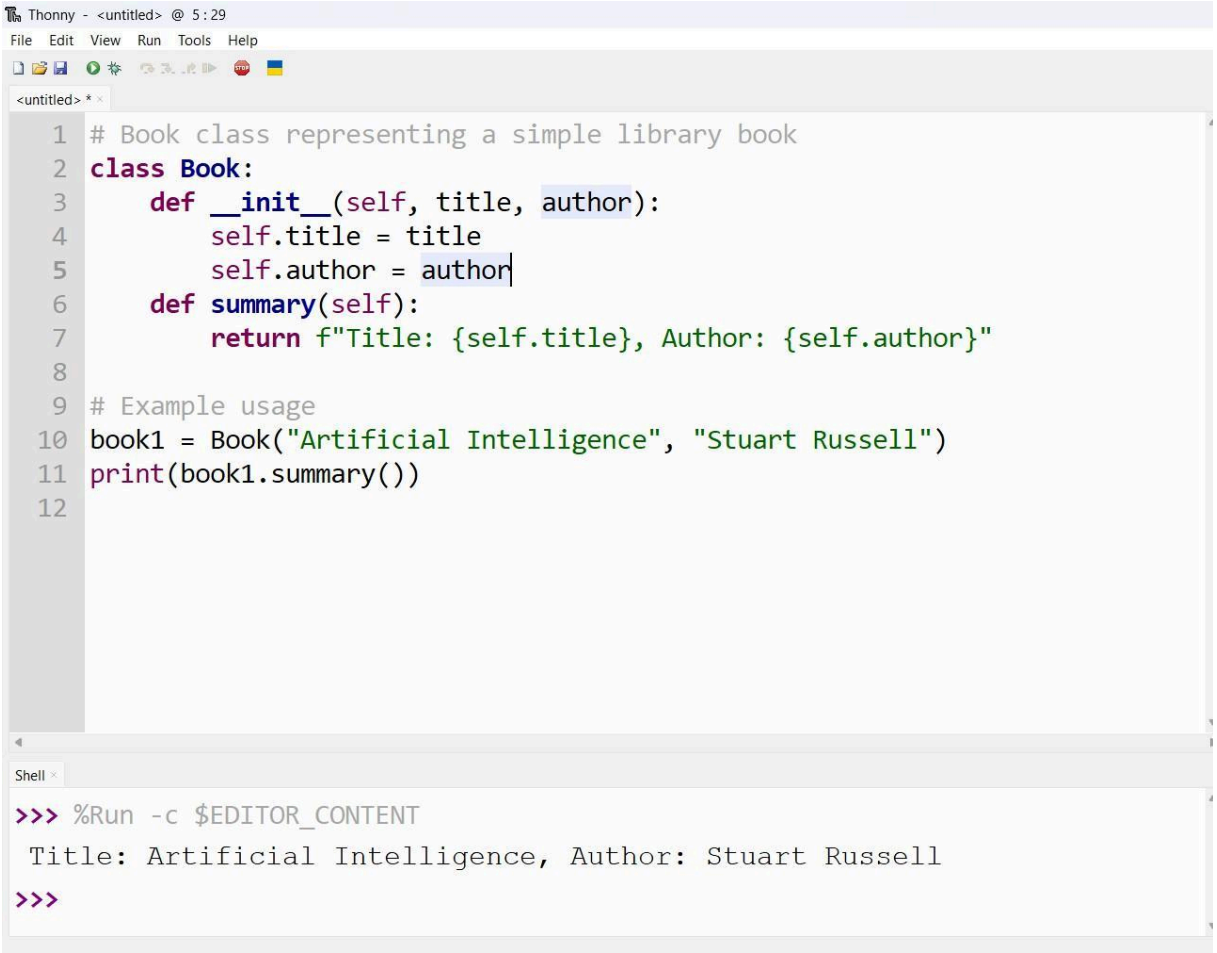
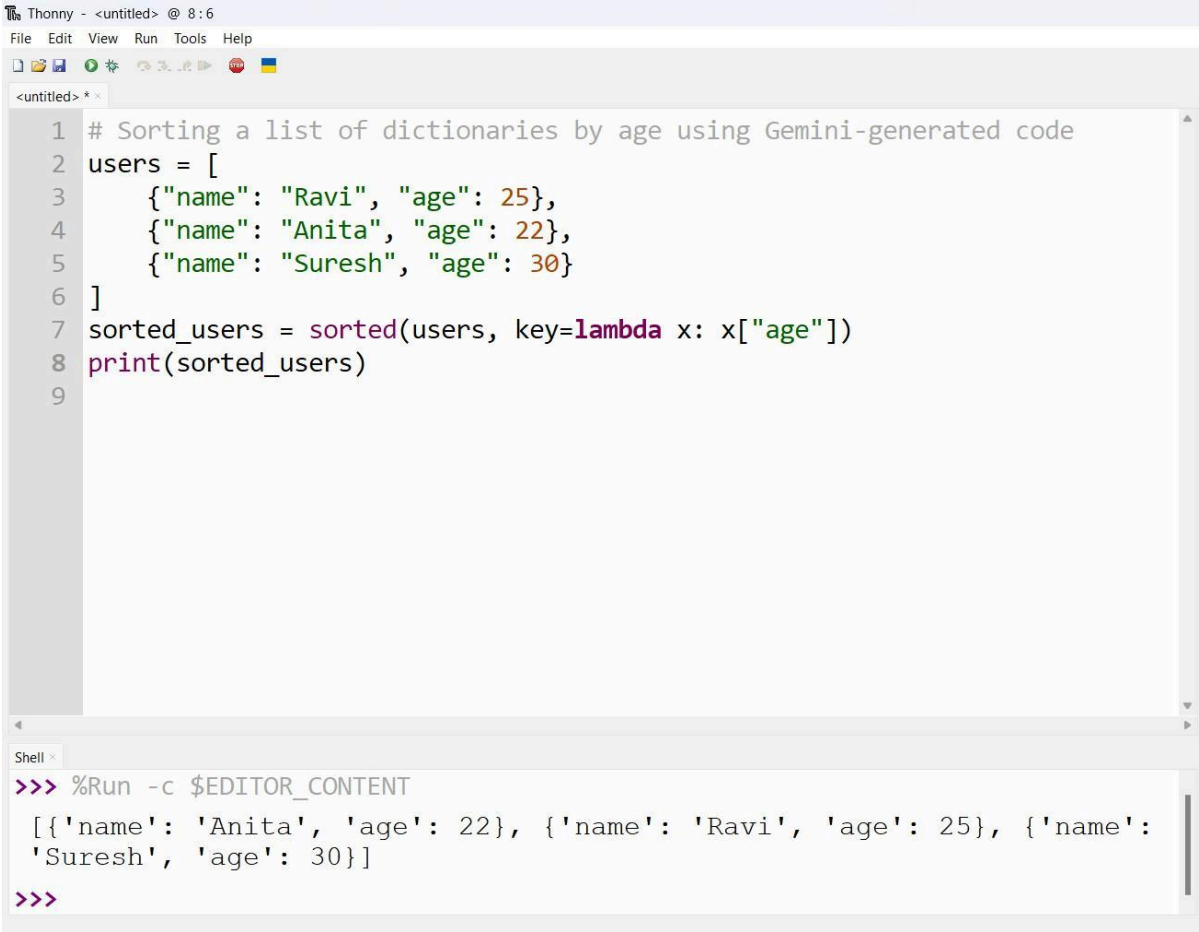
Task 1: Book Class Generation (Using Cursor AI)

Python Code: Book Class

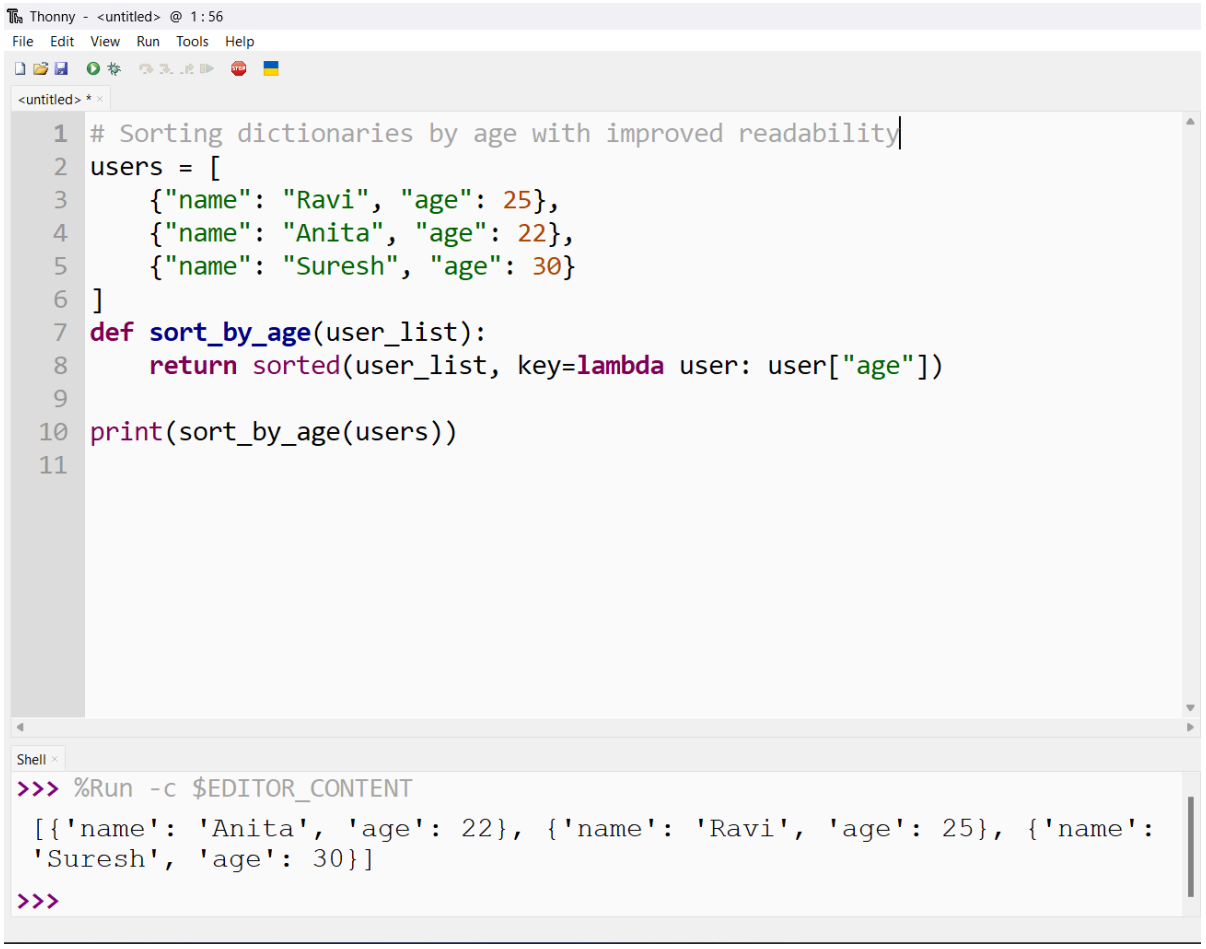


Task 2: Sorting Dictionaries with AI

- **Using Google Gemini**



Using Cursor AI



Comparison (Short Note):
The solution generated by Gemini is short and direct, making it quick to understand for simple tasks. Cursor AI, on the other hand, focused more on clean structure by using a function, which improves readability and makes the code reusable. While both approaches have similar performance, the Cursor AI version is more suitable for larger projects where maintainability and clarity are important.

Task 3: Calculator Using Functions (Gemini)

Calculator Code

Thonny - <untitled> @ 14 : 40

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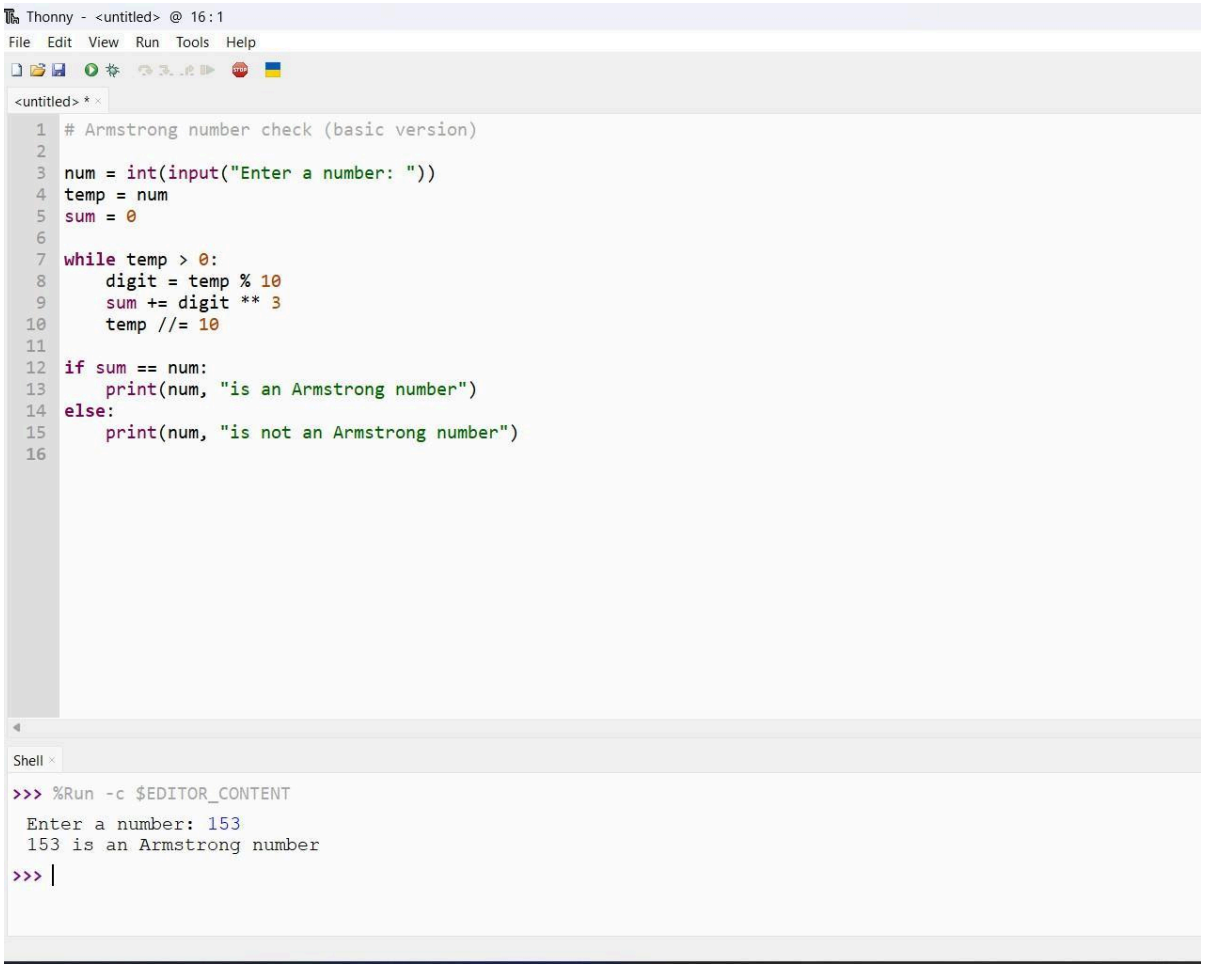
```python
# Basic calculator using functions
def add(a, b):
    return a + b
def subtract(a, b):
    return a - b
def multiply(a, b):
    return a * b
def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed"
    return a / b
# Example usage
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
print("Addition:", add(x, y))
print("Subtraction:", subtract(x, y))
print("Multiplication:", multiply(x, y))
print("Division:", divide(x, y))
```

Shell

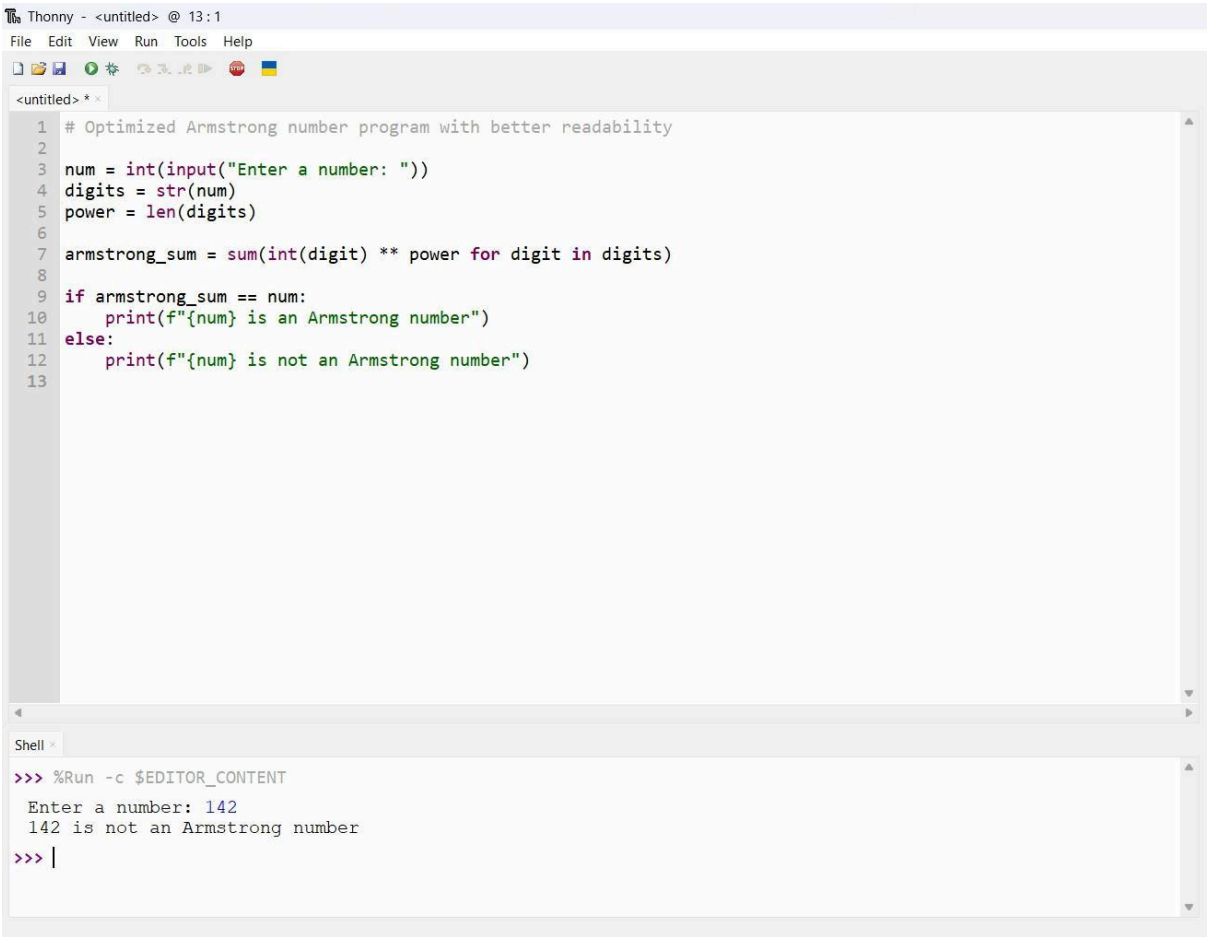
```
>>> %Run -c $EDITOR_CONTENT
 Enter first number: 5
 Enter second number: 18
 Addition: 23
 Subtraction: -13
 Multiplication: 90
 Division: 0.2777777777777778
```

Task 4: Armstrong Number Optimization

◆ **Version 1: Gemini-Generated Armstrong Program**



Version 2: Optimized Using Cursor AI



Summary of Improvements:

- **Reduced lines of code**
- **Removed unnecessary variables**
- **Improved readability using Python built-in functions**
- **Works for Armstrong numbers of any length**